TIC 43712813

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        1.705590 d
  Epoch (t0)    1438.0808 BTJD
  Duration      0.0840 d
  Depth         6.118 %  (61,182 ppm)
  SNR           277.0

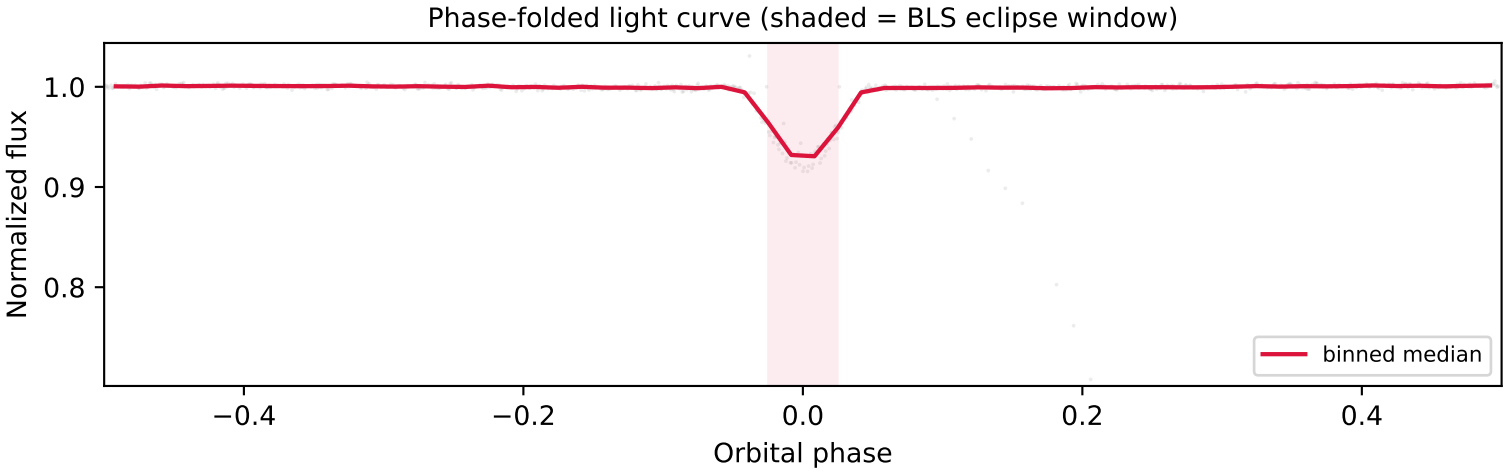
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      15  (gate >= 3)
  Robust depth ratio     1.05  (gate >= 0.50)
  Robust depth sigma     220.8  (gate >= 7.0)
  Odd/even z-score       0.93  (consistent)
  Secondary eclipse      not detected
```

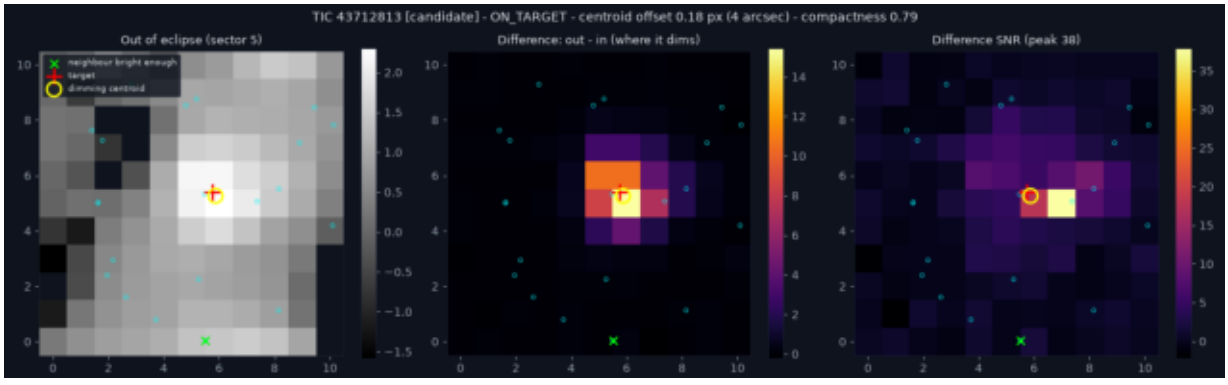
```
CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 5    light-curve dip in this sector: 211.4 sigma    t0 refined by 0.0050 phase
  Difference-image peak SNR   37.91   (gate >= 5.0)      Compactness   0.79   (gate >= 0.50)
  Dimming centroid offset     0.18 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 43712827
    5.24 px away, dTmag +2.69 - farther from the dimming than the target
```





No caveats - clean detection on every check.